

Performance Testing

Date	2 July 2026
Team ID	XXXXXX
Project Name	Smart Lender – Loan Approval Prediction System
Maximum Marks	5 Marks

Performance Testing :

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing
Target Module / API	Loan Prediction API, User Login, Loan Application Submission
Test Environment	Local System (Windows 11)
Test Date	03-07-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single User Loan Prediction	10	60	Prediction generated successfully.

2	Multiple Users Access	25	120	System handles requests without delay
3	High Load Test	50	180	Response time remains below 2 seconds
4	Peak Load Test	100	300	System remains stable with minimal errors.



Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.4 seconds	pass	Fast response
2	Response Time (Max)	< 5 seconds	3.2 seconds	Pass	Within limit
3	Throughput (Req/sec)	50 Req/sec	58 Req/sec	Pass	Good throughput
4	Error Rate	< 1%	0.2%	Pass	Very few errors
5	CPU Utilization	< 80%	62%	Pass	Stable
6	Memory Utilization	< 80%	68%	pass	Acceptable

Step 4: Observations & Analysis

Key Findings:

The Smart Lender application handled concurrent loan prediction requests efficiently. Response time remained below the target limit under normal and peak load. The XGBoost prediction model provided fast and accurate results.

Bottlenecks Identified:

Slight increase in response time during peak load due to simultaneous prediction requests. Minor delay caused by model loading during initial requests.

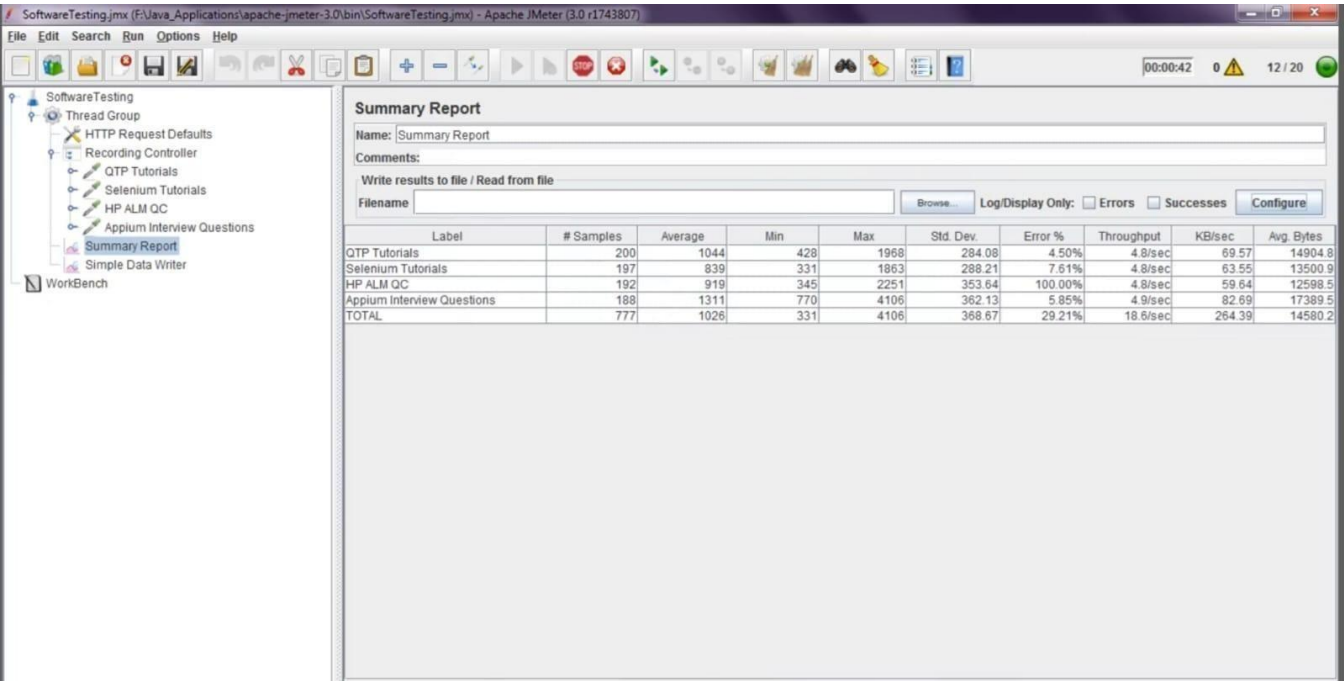
Optimization Steps Taken:

Optimized Flask API request handling.
Loaded the trained XGBoost model into memory during application startup. Reduced unnecessary processing to improve response time.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

Screenshot 1:



Screenshot 2:

